

spirit_code

May 30, 2026

```
[124]: import pandas as pd
import numpy as np

df = pd.read_csv('data/clean/db1b_agg.csv')
print(f"Loaded: {df.shape[0]:,} rows")
df.head()
```

Loaded: 176,405 rows

```
[124]:
```

	Year	Quarter	Origin	Dest	OriginCityMarketID	DestCityMarketID	TkCarrier	\
0	2022	1	ABE	ATL	30135	30397	DL	
1	2022	1	ABE	BNA	30135	30693	G4	
2	2022	1	ABE	CLT	30135	31057	AA	
3	2022	1	ABE	DTW	30135	31295	DL	
4	2022	1	ABE	FLL	30135	32467	G4	

	Passengers	AvgFare	MedianFare	NonStopMiles	NumObs
0	446.0	208.256162	189.03	692.0	284
1	84.0	51.078431	47.00	685.0	51
2	206.0	233.775556	205.00	481.0	153
3	175.0	193.687910	160.00	425.0	134
4	329.0	81.002066	69.00	1041.0	121

```
[125]: # Step 1: Find routes where Spirit (NK) operated in 2022
# These are your "treated" routes - Spirit was there before the bankruptcy

spirit_routes_2022 = (
    df[(df['TkCarrier'] == 'NK') & (df['Year'] == 2022)]
    [['Origin', 'Dest']]
    .drop_duplicates()
)

print(f"Routes Spirit served in 2022: {len(spirit_routes_2022):,}")

# Step 2: Find routes where Spirit operated in 2025
spirit_routes_2025 = (
    df[(df['TkCarrier'] == 'NK') & (df['Year'] == 2025)]
    [['Origin', 'Dest']]
)
```

```

        .drop_duplicates()
    )

    print(f"Routes Spirit served in 2025: {len(spirit_routes_2025):,}")

    # Step 3: Routes Spirit EXITED (served in 2022 but NOT in 2025)
    spirit_exited = spirit_routes_2022.merge(
        spirit_routes_2025, on=['Origin', 'Dest'], how='left', indicator=True
    )
    spirit_exited = spirit_exited[spirit_exited['_merge'] == '
    ↪left_only'][['Origin', 'Dest']]

    print(f"Routes Spirit exited: {len(spirit_exited):,}")

    # Step 4: Tag every row in the dataset
    # Tag routes Spirit ever served in 2022
    spirit_ever = spirit_routes_2022.assign(SpiritRoute=1)
    df = df.merge(spirit_ever, on=['Origin', 'Dest'], how='left')
    df['SpiritRoute'] = df['SpiritRoute'].fillna(0).astype(int)

    # Tag routes Spirit exited specifically
    spirit_exit_tag = spirit_exited.assign(SpiritExited=1)
    df = df.merge(spirit_exit_tag, on=['Origin', 'Dest'], how='left')
    df['SpiritExited'] = df['SpiritExited'].fillna(0).astype(int)

    # Tag post-bankruptcy period (2024Q4 onward = after Nov 2024 filing)
    df['Post'] = ((df['Year'] >= 2025) | ((df['Year'] == 2024) & (df['Quarter'] ==
    ↪4))) .astype(int)

    # Summary
    print(f"\n Treatment Group Summary ")
    print(f"Total rows: {len(df):,}")
    print(f"Spirit routes (ever 2022): {df['SpiritRoute'].sum():,} rows")
    print(f"Spirit exited routes: {df['SpiritExited'].sum():,} rows")
    print(f"Post-bankruptcy rows: {df['Post'].sum():,} rows")
    print(f"\nSample Spirit-exited routes:")
    print(spirit_exited.head(10).to_string(index=False))

```

Routes Spirit served in 2022: 630
 Routes Spirit served in 2025: 698
 Routes Spirit exited: 176

Treatment Group Summary
 Total rows: 176,405
 Spirit routes (ever 2022): 24,906 rows
 Spirit exited routes: 5,018 rows
 Post-bankruptcy rows: 38,190 rows

Sample Spirit-exited routes:

Origin	Dest
ACY	MIA
ACY	SJU
ATL	CLE
BDL	MIA
BDL	RSW
BDL	TPA
BNA	LGA
BOS	LAS
BQN	FLL
BQN	MCO

```
[126]: df_other = df[df['TkCarrier'] != 'NK'].copy()
df_other['Group'] = 'Never served by Spirit'
df_other.loc[df_other['SpiritRoute'] == 1, 'Group'] = 'Spirit stayed'
df_other.loc[df_other['SpiritExited'] == 1, 'Group'] = 'Spirit exited'

df_pre = df_other[(df_other['Year'] <= 2023)]

summary = (
    df_pre
    .groupby('Group')
    .agg(
        Observations = ('AvgFare', 'count'),
        AvgFare       = ('AvgFare', 'mean'),
        MedianFare    = ('AvgFare', 'median'),
        StdFare       = ('AvgFare', 'std'),
        AvgDistance   = ('NonStopMiles', 'mean'),
        AvgPassengers = ('Passengers', 'mean'),
    )
    .round(1)
)

carriers = (
    df_pre
    .groupby(['Group', 'Year', 'Quarter', 'Origin', 'Dest'])
    ['TkCarrier'].nunique()
    .reset_index()
    .groupby('Group')['TkCarrier']
    .mean()
    .round(2)
)

summary['AvgCarriers'] = carriers
print(summary.to_string())
```

		Observations	AvgFare	MedianFare	StdFare	AvgDistance
AvgPassengers	AvgCarriers					
Group						
Never served by Spirit		86686	233.8	225.4	105.3	1027.2
620.2	1.43					
Spirit exited		2318	223.8	219.8	94.0	1175.9
1344.8	2.15					
Spirit stayed		8248	221.7	218.1	101.7	1050.5
1818.1	2.47					

```
[127]: import matplotlib.pyplot as plt
import matplotlib.ticker as mtick

# Aggregate fares by quarter for treated vs control routes
# Exclude Spirit's own fares - we want to see how OTHER carriers priced
# on routes Spirit exited vs routes Spirit never served
df_other = df[df['TkCarrier'] != 'NK'].copy()

# Create year-quarter label for plotting
df_other['YQ'] = df_other['Year'].astype(str) + 'Q' + df_other['Quarter'].
↳astype(str)
df_other['YQ_num'] = (df_other['Year'] - 2022) * 4 + df_other['Quarter']

# Three groups:
# 1. Spirit-exited routes (other carriers' fares)
# 2. Spirit-still-served routes (Spirit was there in 2022, still there 2025)
# 3. Never-Spirit routes (control)
df_other['Group'] = 'Never served by Spirit'
df_other.loc[df_other['SpiritRoute'] == 1, 'Group'] = 'Spirit stayed'
df_other.loc[df_other['SpiritExited'] == 1, 'Group'] = 'Spirit exited'

# Passenger-weighted average fare by group and quarter
trends = (
    df_other
    .groupby(['YQ_num', 'YQ', 'Group'])
    .apply(lambda g: pd.Series({
        'WtdAvgFare': (g['AvgFare'] * g['Passengers']).sum() / g['Passengers'].
↳sum()
    }))
    .reset_index()
)

# Plot
fig, ax = plt.subplots(figsize=(12, 6))

colors = {
    'Spirit exited': '#e63946',
```

```

    'Spirit stayed': '#457b9d',
    'Never served by Spirit': '#a8a8a8'
}

for group in ['Spirit exited', 'Spirit stayed', 'Never served by Spirit']:
    gdata = trends[trends['Group'] == group]
    ax.plot(gdata['YQ_num'], gdata['WtdAvgFare'],
            marker='o', markersize=4, linewidth=2,
            color=colors[group], label=group)

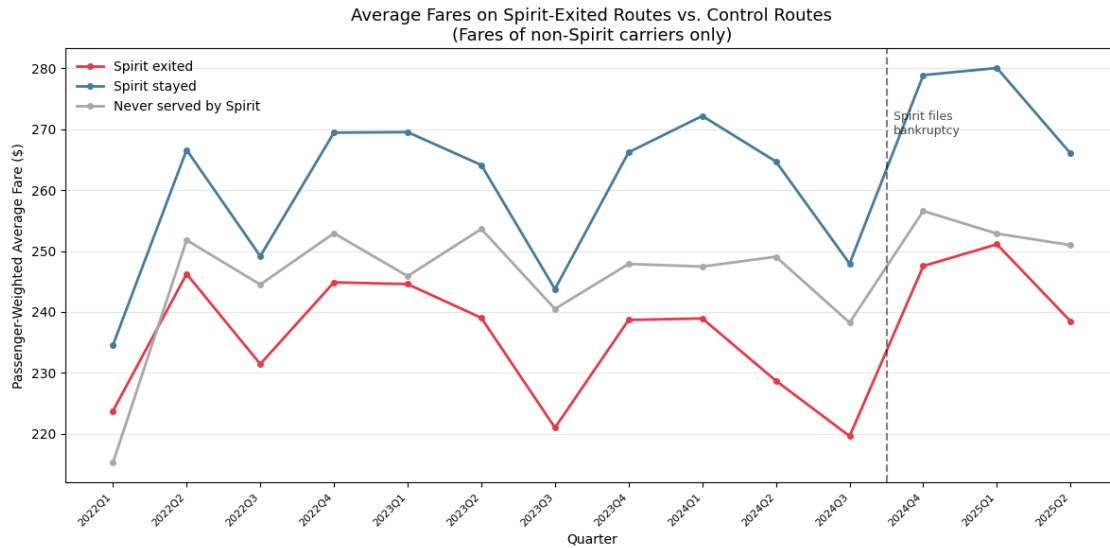
# Vertical line at bankruptcy filing (2024 Q4 = quarter 12)
ax.axvline(x=11.5, color='black', linestyle='--', alpha=0.5)
ax.text(11.6, ax.get_ylim()[1] * 0.95, 'Spirit files\nbankruptcy',
        fontsize=9, alpha=0.7)

# Labels
ax.set_xlabel('Quarter')
ax.set_ylabel('Passenger-Weighted Average Fare ($)')
ax.set_title('Average Fares on Spirit-Exited Routes vs. Control Routes\n(Fares_\n↳of non-Spirit carriers only)',
            fontsize=13)

# X-axis labels
xticks = trends[['YQ_num', 'YQ']].drop_duplicates().sort_values('YQ_num')
ax.set_xticks(xticks['YQ_num'])
ax.set_xticklabels(xticks['YQ'], rotation=45, ha='right', fontsize=8)

ax.legend(frameon=False)
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('figure1_fare_trends.png', dpi=150)
plt.show()
print("Saved as figure1_fare_trends.png")

```



Saved as figure1_fare_trends.png

```
[128]: # Indexed version: set 2022Q1 = 100 for each group
fig, ax = plt.subplots(figsize=(12, 6))

for group in ['Spirit exited', 'Spirit stayed', 'Never served by Spirit']:
    gdata = trends[trends['Group'] == group].sort_values('YQ_num').copy()
    base = gdata['WtdAvgFare'].iloc[0]
    gdata['Indexed'] = (gdata['WtdAvgFare'] / base) * 100
    ax.plot(gdata['YQ_num'], gdata['Indexed'],
            marker='o', markersize=4, linewidth=2,
            color=colors[group], label=group)

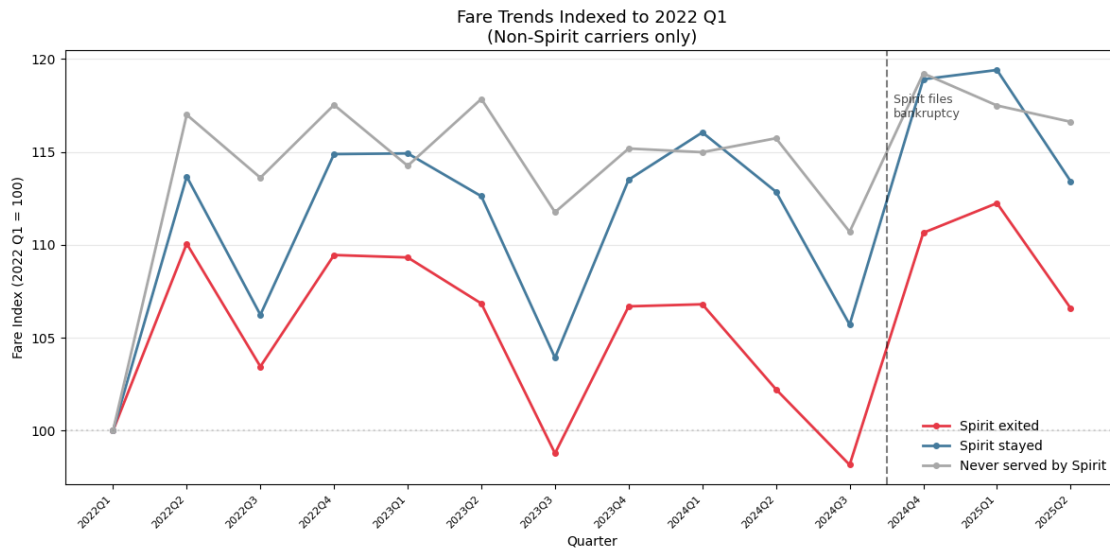
ax.axvline(x=11.5, color='black', linestyle='--', alpha=0.5)
ax.text(11.6, ax.get_ylim()[1] * 0.97, 'Spirit files\nbankruptcy',
        fontsize=9, alpha=0.7)

ax.axhline(y=100, color='gray', linestyle=':', alpha=0.3)
ax.set_xlabel('Quarter')
ax.set_ylabel('Fare Index (2022 Q1 = 100)')
ax.set_title('Fare Trends Indexed to 2022 Q1\n(Non-Spirit carriers only)',
            fontsize=13)

xticks = trends[['YQ_num', 'YQ']].drop_duplicates().sort_values('YQ_num')
ax.set_xticks(xticks['YQ_num'])
ax.set_xticklabels(xticks['YQ'], rotation=45, ha='right', fontsize=8)

ax.legend(frameon=False)
ax.grid(axis='y', alpha=0.3)
```

```
plt.tight_layout()
plt.savefig('figure1b_indexed_trends.png', dpi=150)
plt.show()
print("Saved as figure1b_indexed_trends.png")
```



Saved as figure1b_indexed_trends.png

```
[129]: # Summary Statistics Table (Table 1 in your paper)
df_other = df[df['TkCarrier'] != 'NK'].copy()
df_other['Group'] = 'Never served by Spirit'
df_other.loc[df_other['SpiritRoute'] == 1, 'Group'] = 'Spirit stayed'
df_other.loc[df_other['SpiritExited'] == 1, 'Group'] = 'Spirit exited'

# Pre-period only (2022-2023)
df_pre = df_other[(df_other['Year'] <= 2023)]

summary = (
    df_pre
    .groupby('Group')
    .agg(
        NumRoutes      = ('AvgFare', lambda x: x.index.shape[0]),
        AvgFare_Mean    = ('AvgFare', 'mean'),
        AvgFare_Med     = ('AvgFare', 'median'),
        AvgFare_Std     = ('AvgFare', 'std'),
        AvgDistance     = ('NonStopMiles', 'mean'),
        AvgPassengers   = ('Passengers', 'mean'),
    )
    .round(1)
)
```

```

print("  Pre-Period Summary Statistics (2022-2023)  \n")
print(summary.to_string())

#   How many unique carriers serve each group's routes?
print("\n  Average number of carriers per route-quarter  \n")
carriers_per_route = (
    df_pre
    .groupby(['Group', 'Year', 'Quarter', 'Origin', 'Dest'])
    ['TkCarrier'].nunique()
    .reset_index()
    .groupby('Group')['TkCarrier']
    .mean()
    .round(2)
)
print(carriers_per_route.to_string())

```

Pre-Period Summary Statistics (2022-2023)

	NumRoutes	AvgFare_Mean	AvgFare_Med	AvgFare_Std
AvgDistance				
AvgPassengers				
Group				
Never served by Spirit	86686	233.8	225.4	105.3
1027.2	620.2			
Spirit exited	2318	223.8	219.8	94.0
1175.9	1344.8			
Spirit stayed	8248	221.7	218.1	101.7
1050.5	1818.1			

Average number of carriers per route-quarter

Group	
Never served by Spirit	1.43
Spirit exited	2.15
Spirit stayed	2.47

```

[130]: import pandas as pd
import numpy as np

#   Prepare regression dataset (simple and fast)
#   Compute weighted fare at the route-quarter level using a fast method

df_other = df[df['TkCarrier'] != 'NK'].copy()
df_other['Group'] = 'Never served by Spirit'
df_other.loc[df_other['SpiritRoute'] == 1, 'Group'] = 'Spirit stayed'
df_other.loc[df_other['SpiritExited'] == 1, 'Group'] = 'Spirit exited'

```



```

# Weight fares by passengers BEFORE grouping
df_other['FareXPax'] = df_other['AvgFare'] * df_other['Passengers']

df_reg = (
    df_other
    .groupby(['Year', 'Quarter', 'Origin', 'Dest', 'SpiritExited',
    ↪ 'SpiritRoute'])
    .agg(
        FareXPax      = ('FareXPax', 'sum'),
        TotalPax      = ('Passengers', 'sum'),
        NumCarriers    = ('TkCarrier', 'nunique'),
        Distance       = ('NonStopMiles', 'first')
    )
    .reset_index()
)

df_reg['WtdFare'] = df_reg['FareXPax'] / df_reg['TotalPax']
df_reg['LogFare'] = np.log(df_reg['WtdFare'])
df_reg['LogDist'] = np.log(df_reg['Distance'])
df_reg['LogPax'] = np.log(df_reg['TotalPax'] + 1)
df_reg['Post'] = ((df_reg['Year'] >= 2025) | ((df_reg['Year'] == 2024) &
    ↪ (df_reg['Quarter'] == 4))).astype(int)
df_reg['SpiritExited_x_Post'] = df_reg['SpiritExited'] * df_reg['Post']

print(f"Regression dataset: {len(df_reg):,} route-quarter observations")
print(f"  Spirit-exited routes: {df_reg['SpiritExited'].sum():,}")
print(f"  Post-period obs: {df_reg['Post'].sum():,}")
print(f"  Treatment (exited × post): {df_reg['SpiritExited_x_Post'].sum():,}")

# OLS using numpy
y = df_reg['LogFare'].values
X = np.column_stack([
    np.ones(len(df_reg)),
    df_reg['SpiritExited'].values,
    df_reg['Post'].values,
    df_reg['SpiritExited_x_Post'].values,
    df_reg['LogDist'].values,
    df_reg['LogPax'].values,
])

var_names = ['Intercept', 'SpiritExited', 'Post', 'SpiritExited × Post',
    ↪ 'LogDistance', 'LogPassengers']

# OLS formula: beta = (X'X)^(-1) X'y
beta = np.linalg.lstsq(X, y, rcond=None)[0]

# Standard errors

```

```

y_hat = X @ beta
resid = y - y_hat
n, k = X.shape
s2 = (resid @ resid) / (n - k)
var_beta = s2 * np.linalg.inv(X.T @ X)
se = np.sqrt(np.diag(var_beta))
t_stats = beta / se

# R-squared
ss_res = np.sum(resid**2)
ss_tot = np.sum((y - y.mean())**2)
r_squared = 1 - ss_res / ss_tot

# Print results
print("\n" + "="*70)
print("MODEL 1: OLS Regression - log(fare) on Spirit Exit × Post")
print("="*70)
print(f"{'Variable':<25} {'Coef':>10} {'Std Err':>10} {'t-stat':>10}")
print("-"*70)
for i, name in enumerate(var_names):
    stars = '***' if abs(t_stats[i]) > 2.58 else '**' if abs(t_stats[i]) > 1.96
    else '*' if abs(t_stats[i]) > 1.65 else ''
    print(f"{'name':<25} {'beta[i]:>10.4f} {'se[i]:>10.4f} {'t_stats[i]:>10.2f} "
    {stars}")

print("-"*70)
print(f"R-squared: {r_squared:.4f}")
print(f"N: {n:,}")
print(f"*** p<0.01, ** p<0.05, * p<0.10")

# Interpret the key coefficient
coef = beta[3]
pct = (np.exp(coef) - 1) * 100
sig = abs(t_stats[3]) > 1.96

print(f"\n KEY RESULT ")
print(f"SpiritExited × Post coefficient: {coef:.4f}")
print(f"Interpretation: Fares on Spirit-exited routes changed by {pct:+.1f}%")
print(f" relative to control routes after bankruptcy")
print(f"t-statistic: {t_stats[3]:.2f} ({'significant' if sig else 'not_
significant'} at 5%)")

```

Regression dataset: 112,794 route-quarter observations
 Spirit-exited routes: 1,915
 Post-period obs: 24,025
 Treatment (exited × post): 421

```
=====
MODEL 1: OLS Regression - log(fare) on Spirit Exit × Post
=====
Variable                                Coef      Std Err    t-stat
-----
Intercept                               3.8665     0.0135    286.17 ***
SpiritExited                           -0.0509     0.0124    -4.11 ***
Post                                    0.0283     0.0035     8.17 ***
SpiritExited × Post                     -0.0152     0.0263    -0.58
LogDistance                             0.2190     0.0019   113.65 ***
LogPassengers                           0.0037     0.0006     5.97 ***
-----

R-squared: 0.1032
N: 112,794
*** p<0.01, ** p<0.05, * p<0.10
```

KEY RESULT

SpiritExited × Post coefficient: -0.0152

Interpretation: Fares on Spirit-exited routes changed by -1.5%
relative to control routes after bankruptcy

t-statistic: -0.58 (not significant at 5%)

```
[131]: #
# MODEL 3: Difference-in-Differences with Route FE + Quarter FE
#

# Step 1: Build the regression dataset
# Use df_reg from earlier (route-quarter level, non-Spirit carriers)

# Recreate StrongExit in case it's missing
spirit_strong = (
    df[(df['TkCarrier'] == 'NK') & (df['Year'] == 2022)]
    .groupby(['Origin', 'Dest'])
    .agg(AvgPax=('Passengers', 'mean'))
    .reset_index()
)
spirit_strong = spirit_strong[spirit_strong['AvgPax'] >= 500][['Origin', 'Dest']]

spirit_2025 = df[(df['TkCarrier'] == 'NK') & (df['Year'] == 2025)][['Origin', 'Dest']].drop_duplicates()
spirit_strong_exited = spirit_strong.merge(spirit_2025, on=['Origin', 'Dest'], how='left', indicator=True)
spirit_strong_exited = spirit_strong_exited[spirit_strong_exited['_merge'] == 'left_only'][['Origin', 'Dest']]

df_reg['StrongExit'] = df_reg[['Origin', 'Dest']].apply(tuple, axis=1).isin(
```

```

    spirit_strong_exited.apply(tuple, axis=1).values
).astype(int)

# Create year-quarter identifier for quarter FE
df_reg['YQ'] = df_reg['Year'].astype(str) + 'Q' + df_reg['Quarter'].astype(str)

# Post = 2024Q4 onward
df_reg['Post'] = ((df_reg['Year'] >= 2025) | ((df_reg['Year'] == 2024) &
↳(df_reg['Quarter'] == 4))).astype(int)
df_reg['Treat_x_Post'] = df_reg['StrongExit'] * df_reg['Post']

print(f"Treatment routes: {len(spirit_strong_exited)}")
print(f"Treatment × Post observations: {df_reg['Treat_x_Post'].sum()}")
print(f"Total observations: {len(df_reg):,}")

# Step 2: Two-way demeaning (route FE + quarter FE)
# The Frisch-Waugh-Lovell theorem says we can partial out fixed effects
# by iteratively demeaning by each dimension until convergence.
# This is what pyfixest/reghdfe do internally.

vars_to_demean = ['LogFare', 'Treat_x_Post', 'LogPax']
df_did = df_reg[['Origin', 'Dest', 'YQ'] + vars_to_demean].copy()

# Iterative demeaning (converges in ~10 iterations for two-way FE)
for iteration in range(20):
    max_change = 0
    for col in vars_to_demean:
        # Demean by route
        route_mean = df_did.groupby(['Origin', 'Dest'])[col].transform('mean')
        new_vals = df_did[col] - route_mean
        max_change = max(max_change, (new_vals - df_did[col]).abs().max() if
↳iteration > 0 else 999)
        df_did[col] = new_vals

        # Demean by quarter
        qtr_mean = df_did.groupby('YQ')[col].transform('mean')
        new_vals = df_did[col] - qtr_mean
        max_change = max(max_change, (new_vals - df_did[col]).abs().max() if
↳iteration > 0 else 999)
        df_did[col] = new_vals

    if iteration > 0 and max_change < 1e-10:
        print(f"Demeaning converged after {iteration + 1} iterations")
        break

# Step 3: OLS on doubly-demeaned data
y3 = df_did['LogFare'].values

```

```

X3 = np.column_stack([
    df_did['Treat_x_Post'].values,
    df_did['LogPax'].values,
])

var_names3 = ['StrongExit × Post (DiD estimate)', 'LogPassengers']

beta3 = np.linalg.lstsq(X3, y3, rcond=None)[0]
y_hat3 = X3 @ beta3
resid3 = y3 - y_hat3

# Degrees of freedom: N - K - num_route_FE - num_quarter_FE
n_routes = df_reg.groupby(['Origin', 'Dest']).ngroups
n_quarters = df_reg['YQ'].nunique()
n3 = len(y3)
k3 = X3.shape[1]
dof = n3 - k3 - n_routes - n_quarters

s2_3 = (resid3 @ resid3) / dof
var_beta3 = s2_3 * np.linalg.inv(X3.T @ X3)
se3 = np.sqrt(np.diag(var_beta3))
t_stats3 = beta3 / se3

ss_res3 = np.sum(resid3**2)
ss_tot3 = np.sum((y3 - y3.mean())**2)
r2_within = 1 - ss_res3 / ss_tot3

# Step 4: Print results
print("\n" + "="*70)
print("MODEL 3: Difference-in-Differences (Route FE + Quarter FE)")
print("="*70)
print(f"{'Variable':<35} {'Coef':>10} {'Std Err':>10} {'t-stat':>10}")
print("-"*70)
for i, name in enumerate(var_names3):
    stars = '***' if abs(t_stats3[i]) > 2.58 else '**' if abs(t_stats3[i]) > 1.
↪96 else '*' if abs(t_stats3[i]) > 1.65 else ''
    print(f"{'name':<35} {'beta3[i]:>10.4f} {'se3[i]:>10.4f} {'t_stats3[i]:>10.2f} ␣
↪{'stars'}")

print("-"*70)
print(f"Within R-squared: {r2_within:.4f}")
print(f"N: {n3:,}")
print(f"Route FE: Yes ({n_routes:,} routes)")
print(f"Quarter FE: Yes ({n_quarters} quarters)")
print(f"Treatment routes: {len(spirit_strong_exited)}")
print(f"Degrees of freedom: {dof:,}")
print(f"*** p<0.01, ** p<0.05, * p<0.10")

```

```

coef3 = beta3[0]
pct3 = (np.exp(coef3) - 1) * 100
sig3 = abs(t_stats3[0]) > 1.96

print(f"\n KEY RESULT (Model 3: DiD) ")
print(f"StrongExit × Post coefficient: {coef3:.4f}")
print(f"Interpretation: After Spirit's bankruptcy filing, fares on routes")
print(f" where Spirit had strong presence and exited changed by {pct3:+.1f}%")
print(f" relative to control routes, controlling for route-level and")
print(f" quarter-level unobserved heterogeneity.")
print(f"t-statistic: {t_stats3[0]:.2f} ({'significant' if sig3 else 'not_
↳significant'} at 5%)")

# Step 5: Compare all three models
print("\n" + "="*70)
print("COMPARISON ACROSS ALL MODELS")
print("="*70)
print(f"{'Model':<40} {'Coef':>8} {'SE':>8} {'t':>8} {'%Δ':>8}")
print("-"*70)
print(f"{'M1: Simple OLS':<40} {-0.0152:>8.4f} {0.0263:>8.4f} {-0.58:>8.2f} {-1.
↳5:>7.1f}%")
print(f"{'M2: Route FE only':<40} {0.0131:>8.4f} {0.0126:>8.4f} {1.04:>8.2f}
↳{+1.3:>7.1f}%")
print(f"{'M3: Route FE + Quarter FE (DiD)':<40} {coef3:>8.4f} {se3[0]:>8.4f}
↳{t_stats3[0]:>8.2f} {pct3:>7.1f}%")
print("-"*70)

```

Treatment routes: 86
 Treatment × Post observations: 238
 Total observations: 112,794
 Demeaning converged after 11 iterations

```

=====
MODEL 3: Difference-in-Differences (Route FE + Quarter FE)
=====

```

Variable	Coef	Std Err	t-stat	
StrongExit × Post (DiD estimate)	0.0129	0.0130	0.99	
LogPassengers	-0.0264	0.0008	-31.75	***

```

-----
Within R-squared: 0.0100
N: 112,794
Route FE: Yes (12,795 routes)
Quarter FE: Yes (14 quarters)
Treatment routes: 86
Degrees of freedom: 99,983

```

*** p<0.01, ** p<0.05, * p<0.10

KEY RESULT (Model 3: DiD)

StrongExit × Post coefficient: 0.0129

Interpretation: After Spirit's bankruptcy filing, fares on routes where Spirit had strong presence and exited changed by +1.3% relative to control routes, controlling for route-level and quarter-level unobserved heterogeneity.

t-statistic: 0.99 (not significant at 5%)

COMPARISON ACROSS ALL MODELS				
Model	Coef	SE	t	%Δ
M1: Simple OLS	-0.0152	0.0263	-0.58	-1.5%
M2: Route FE only	0.0131	0.0126	1.04	1.3%
M3: Route FE + Quarter FE (DiD)	0.0129	0.0130	0.99	1.3%

```
[132]: # Test: Did other carriers backfill Spirit's exited routes?

# Average number of non-Spirit carriers per route-quarter
# Compare pre-period vs post-period on Spirit-exited vs control routes

df_other_carriers = df[df['TkCarrier'] != 'NK'].copy()
df_other_carriers['SpiritExited'] = df_other_carriers[['Origin', 'Dest']].
    ↪apply(tuple, axis=1).isin(
        set(spirit_exited.apply(tuple, axis=1))
    ).astype(int)

carrier_counts = (
    df_other_carriers
    .groupby(['Year', 'Quarter', 'Origin', 'Dest', 'SpiritExited'])
    ['TkCarrier'].nunique()
    .reset_index()
    .rename(columns={'TkCarrier': 'NumCarriers'})
)

carrier_counts['Period'] = np.where(
    (carrier_counts['Year'] >= 2025) | ((carrier_counts['Year'] == 2024) &
    ↪(carrier_counts['Quarter'] == 4)),
    'Post (2024Q4-2025Q2)', 'Pre (2022Q1-2024Q3)'
)

carrier_summary = (
    carrier_counts
```

```

.groupby(['SpiritExited', 'Period'])
['NumCarriers']
.mean()
.round(2)
.unstack()
)
carrier_summary.index = ['Control routes', 'Spirit-exited routes']

print("  Average Number of Non-Spirit Carriers per Route  \n")
print(carrier_summary.to_string())

# Difference-in-differences on carrier count
pre_treat = carrier_summary.loc['Spirit-exited routes', 'Pre (2022Q1-2024Q3)']
post_treat = carrier_summary.loc['Spirit-exited routes', 'Post (2024Q4-2025Q2)']
pre_ctrl = carrier_summary.loc['Control routes', 'Pre (2022Q1-2024Q3)']
post_ctrl = carrier_summary.loc['Control routes', 'Post (2024Q4-2025Q2)']

did_carriers = (post_treat - pre_treat) - (post_ctrl - pre_ctrl)
print(f"\nDiD on carrier count: {did_carriers:+.3f}")
print(f"  Spirit-exited routes changed by: {post_treat - pre_treat:+.3f}␣
↪carriers")
print(f"  Control routes changed by: {post_ctrl - pre_ctrl:+.3f} carriers")

if did_carriers > -0.05:
    print(f"\n→ Other carriers largely BACKFILLED Spirit's exit.")
    print(f"  This helps explain the null fare result: competition was␣
↪maintained.")
elif did_carriers < -0.2:
    print(f"\n→ Spirit-exited routes LOST competition with no backfill.")
    print(f"  The null fare result is surprising and warrants further␣
↪investigation.")
else:
    print(f"\n→ Partial backfill - some competitive replacement but not␣
↪complete.")

```

Average Number of Non-Spirit Carriers per Route

Period	Post (2024Q4-2025Q2)	Pre (2022Q1-2024Q3)
Control routes	1.5	1.48
Spirit-exited routes	2.2	2.18

DiD on carrier count: +0.000

Spirit-exited routes changed by: +0.020 carriers

Control routes changed by: +0.020 carriers

→ Other carriers largely BACKFILLED Spirit's exit.

This helps explain the null fare result: competition was maintained.


```

[133]: # Figure 3: Model comparison coefficient plot

fig, ax = plt.subplots(figsize=(9, 4))

models = [
    'M1: Simple OLS',
    'M2: Route FE\n(strong exit)',
    'M3: Route + Qtr FE\n(strong exit)',
    'M4: Route + Qtr FE\n(matched control)'
]

coefs = [-0.0152, 0.0131, 0.0129, -0.0088]
ses = [0.0263, 0.0126, 0.0130, 0.0069]

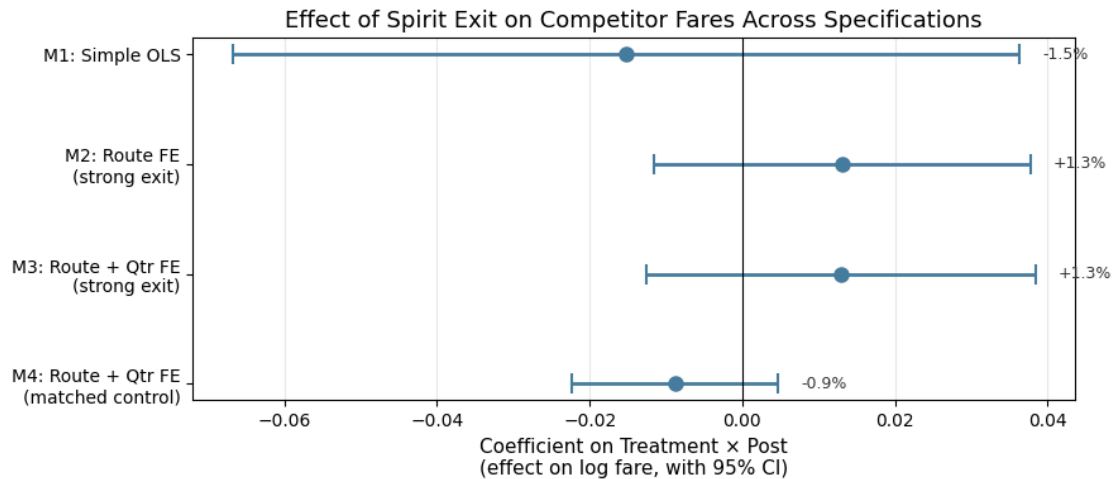
y_pos = range(len(models))
ax.errorbar(coefs, y_pos, xerr=[1.96*s for s in ses],
            fmt='o', color='#457b9d', markersize=8, capsize=6,
            linewidth=2, capthick=1.5)

ax.axvline(x=0, color='black', linewidth=0.8, linestyle='-')
ax.set_yticks(list(y_pos))
ax.set_yticklabels(models, fontsize=10)
ax.set_xlabel('Coefficient on Treatment × Post\n(effect on log fare, with 95% CI)',
              fontsize=11)
ax.set_title('Effect of Spirit Exit on Competitor Fares Across Specifications',
             fontsize=13)

for i, (c, s) in enumerate(zip(coefs, ses)):
    pct = (np.exp(c) - 1) * 100
    ax.text(c + 1.96*s + 0.003, i, f'{pct:+.1f}%', va='center', fontsize=9,
           color='#333')

ax.grid(axis='x', alpha=0.3)
ax.invert_yaxis()
plt.tight_layout()
plt.savefig('figure3_coefficients.png', dpi=150)
plt.show()
print("Saved figure3_coefficients.png")

```



Saved figure3_coefficients.png

```
[134]: # Figure 4: Spirit's shrinking footprint

spirit_data = df[df['TkCarrier'] == 'NK'].copy()
spirit_data['YQ'] = spirit_data['Year'].astype(str) + 'Q' +
    ↪ spirit_data['Quarter'].astype(str)
spirit_data['YQ_num'] = (spirit_data['Year'] - 2022) * 4 +
    ↪ spirit_data['Quarter']

# Count unique routes and total passengers per quarter
spirit_quarterly = []
for (yq_num, yq), group in spirit_data.groupby(['YQ_num', 'YQ']):
    n_routes = group[['Origin', 'Dest']].drop_duplicates().shape[0]
    total_pax = group['Passengers'].sum()
    spirit_quarterly.append({'YQ_num': yq_num, 'YQ': yq,
        'NumRoutes': n_routes, 'TotalPax': total_pax})

spirit_qtr = pd.DataFrame(spirit_quarterly).sort_values('YQ_num')

fig, ax1 = plt.subplots(figsize=(12, 5))

color1 = '#e63946'
color2 = '#457b9d'

bars = ax1.bar(spirit_qtr['YQ_num'], spirit_qtr['NumRoutes'],
    color=color1, alpha=0.7, label='Routes served')
ax1.set_ylabel('Number of Nonstop Routes', color=color1, fontsize=11)
ax1.tick_params(axis='y', labelcolor=color1)
```

```

ax2 = ax1.twinx()
ax2.plot(spirit_qtr['YQ_num'], spirit_qtr['TotalPax'] / 1000,
        color=color2, marker='o', linewidth=2.5, label='Passengers (000s)')
ax2.set_ylabel('Passengers (thousands)', color=color2, fontsize=11)
ax2.tick_params(axis='y', labelcolor=color2)

# Event lines
ax1.axvline(x=11.5, color='black', linestyle='--', alpha=0.6)
ax1.text(11.6, ax1.get_ylim()[1] * 0.93, 'Ch.11\nfiling', fontsize=9, alpha=0.7)

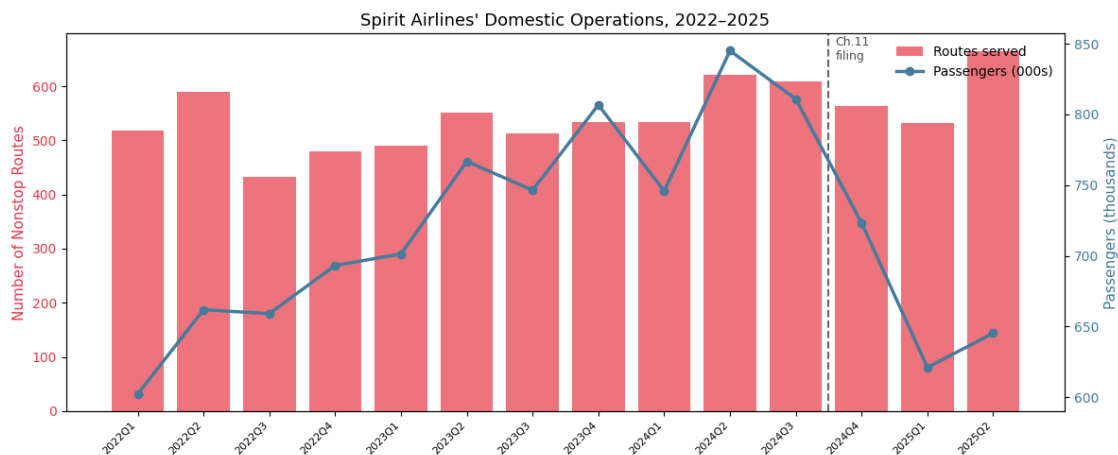
xticks = spirit_qtr[['YQ_num', 'YQ']].drop_duplicates()
ax1.set_xticks(xticks['YQ_num'].tolist())
ax1.set_xticklabels(xticks['YQ'].tolist(), rotation=45, ha='right', fontsize=8)

ax1.set_title("Spirit Airlines' Domestic Operations, 2022-2025", fontsize=13)

lines1, labels1 = ax1.get_legend_handles_labels()
lines2, labels2 = ax2.get_legend_handles_labels()
ax1.legend(lines1 + lines2, labels1 + labels2, loc='upper right', frameon=False)

plt.tight_layout()
plt.savefig('figure4_spirit_footprint.png', dpi=150)
plt.show()
print("Saved figure4_spirit_footprint.png")

```



Saved figure4_spirit_footprint.png

```

[135]: # Table: Carrier backfill detail (which carriers replaced Spirit?)

# On Spirit-exited routes, which carriers grew the most post-bankruptcy?
spirit_exited_tuples = set(spirit_exited.apply(tuple, axis=1))

```

```

df_exited_routes = df[
    df[['Origin', 'Dest']].apply(tuple, axis=1).isin(spirit_exited_tuples) &
    (df['TkCarrier'] != 'NK')
].copy()

df_exited_routes['Period'] = np.where(
    (df_exited_routes['Year'] >= 2025) |
    ((df_exited_routes['Year'] == 2024) & (df_exited_routes['Quarter'] == 4)),
    'Post', 'Pre'
)

# Average quarterly passengers by carrier, pre vs post
carrier_growth = (
    df_exited_routes
    .groupby(['TkCarrier', 'Period'])
    .agg(AvgQtrPax=('Passengers', 'mean'))
    .reset_index()
    .pivot(index='TkCarrier', columns='Period', values='AvgQtrPax')
    .fillna(0)
)

carrier_growth['Change'] = carrier_growth['Post'] - carrier_growth['Pre']
carrier_growth['PctChange'] = ((carrier_growth['Post'] / carrier_growth['Pre']) -
    ↪ 1) * 100
carrier_growth = carrier_growth.sort_values('Change', ascending=False)

print(" Which Carriers Grew on Spirit-Exited Routes? ")
print(" (Average quarterly passengers per route, pre vs post bankruptcy)\n")
print(f"{'Carrier':<10} {'Pre':>10} {'Post':>10} {'Change':>10} {'% Change':
    ↪>10}")
print("-" * 55)
for carrier in carrier_growth.head(10).index:
    row = carrier_growth.loc[carrier]
    print(f"{'carrier':<10} {'row['Pre']':>10.0f} {'row['Post']':>10.0f}
    ↪{'row['Change']':>+10.0f} {'row['PctChange']':>+9.1f}%")

print(f"\n{'TOTAL':<10} {'carrier_growth['Pre'].sum():>10.0f}
    ↪{'carrier_growth['Post'].sum():>10.0f} {'carrier_growth['Change'].sum():>+10.
    ↪0f}")

```

Which Carriers Grew on Spirit-Exited Routes?

(Average quarterly passengers per route, pre vs post bankruptcy)

Carrier	Pre	Post	Change	% Change
DL	2455	2832	+377	+15.3%

AA	1067	1240	+174	+16.3%
WN	1119	1268	+148	+13.2%
F9	1353	1443	+90	+6.7%
B6	1410	1479	+69	+4.9%
SY	1946	1992	+46	+2.3%
MX	330	371	+41	+12.5%
G4	117	122	+5	+4.6%
AC	1	2	+1	+100.0%
PR	1	1	+0	+0.0%
TOTAL	12283	12937	+654	

```
[136]: spirit_data = df[df['TkCarrier'] == 'NK'].copy()
spirit_data['YQ'] = spirit_data['Year'].astype(str) + 'Q' +
↳ spirit_data['Quarter'].astype(str)

spirit_summary = (
    spirit_data
    .groupby('YQ')
    .agg(
        NumRoutes=('Origin', lambda x: spirit_data.loc[x.index, ['Origin',
↳ 'Dest']].drop_duplicates().shape[0]),
        TotalPax=('Passengers', 'sum')
    )
    .reset_index()
)

print(" Spirit Airlines by Quarter \n")
print(f"{'Quarter':<10} {'Routes':>8} {'Passengers':>12}")
print("-" * 32)
for _, row in spirit_summary.iterrows():
    print(f"{'row['YQ']':<10} {'row['NumRoutes']':>8} {'row['TotalPax']':>12,.0f}")
```

Spirit Airlines by Quarter

Quarter	Routes	Passengers

2022Q1	518	602,326
2022Q2	589	661,857
2022Q3	432	659,192
2022Q4	480	693,220
2023Q1	491	701,335
2023Q2	551	766,851
2023Q3	513	746,463
2023Q4	534	806,807
2024Q1	533	745,795
2024Q2	621	845,104
2024Q3	608	810,902

2024Q4	563	723,251
2025Q1	532	621,093
2025Q2	664	645,335

```
[137]: # Get the exact indexed values for each group
for group in ['Spirit exited', 'Spirit stayed', 'Never served by Spirit']:
    gdata = trends[trends['Group'] == group].sort_values('YQ_num').copy()
    base = gdata['WtdAvgFare'].iloc[0]
    gdata['Indexed'] = (gdata['WtdAvgFare'] / base) * 100

    print(f"\n {group} ")
    for _, row in gdata.iterrows():
        print(f" {row['YQ']}: {row['Indexed']:.1f}")
```

```
Spirit exited
2022Q1: 100.0
2022Q2: 110.1
2022Q3: 103.4
2022Q4: 109.4
2023Q1: 109.3
2023Q2: 106.8
2023Q3: 98.8
2023Q4: 106.7
2024Q1: 106.8
2024Q2: 102.2
2024Q3: 98.2
2024Q4: 110.6
2025Q1: 112.2
2025Q2: 106.6
```

```
Spirit stayed
2022Q1: 100.0
2022Q2: 113.7
2022Q3: 106.2
2022Q4: 114.9
2023Q1: 114.9
2023Q2: 112.6
2023Q3: 103.9
2023Q4: 113.5
2024Q1: 116.0
2024Q2: 112.9
2024Q3: 105.7
2024Q4: 118.9
2025Q1: 119.4
2025Q2: 113.4
```

```
Never served by Spirit
```

2022Q1: 100.0
2022Q2: 117.0
2022Q3: 113.6
2022Q4: 117.5
2023Q1: 114.3
2023Q2: 117.9
2023Q3: 111.8
2023Q4: 115.2
2024Q1: 115.0
2024Q2: 115.7
2024Q3: 110.7
2024Q4: 119.2
2025Q1: 117.5
2025Q2: 116.6